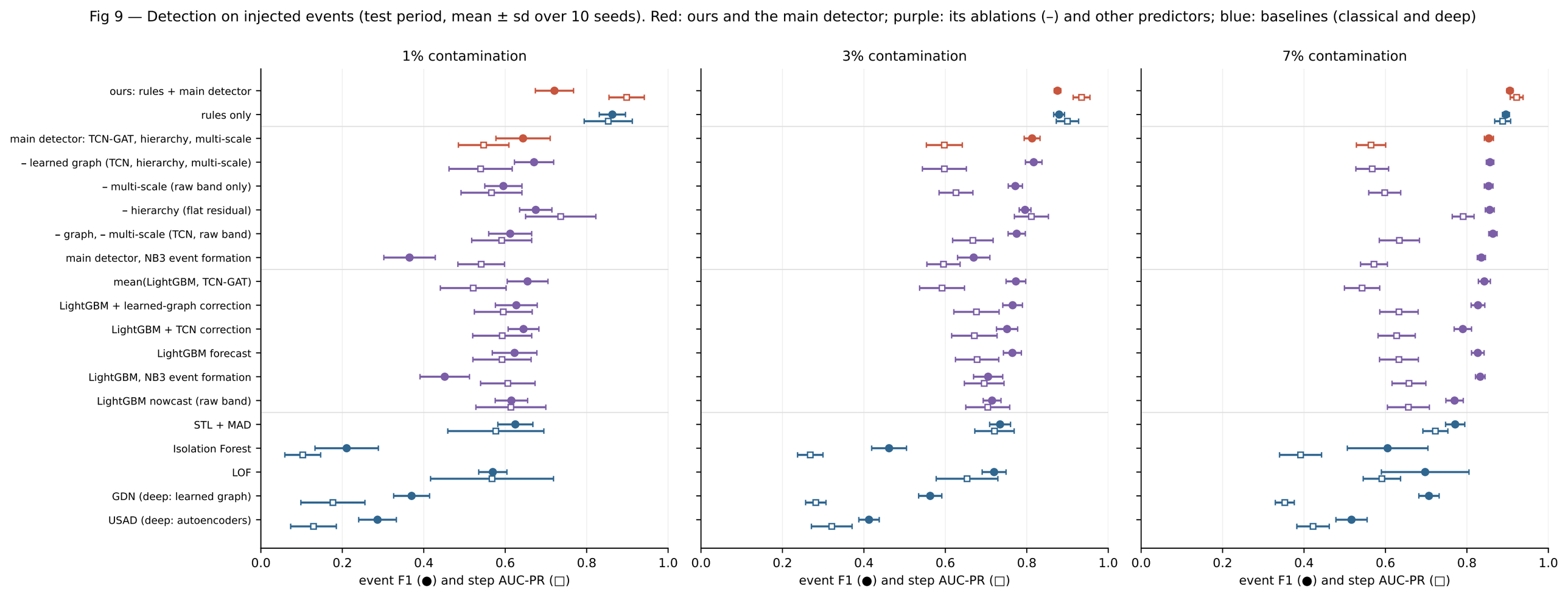
Fig 9 — Detection on injected events (test period, mean ± sd over 10 seeds). Red: ours and the main detector; purple: its ablations (–) and other predictors; blue: baselines (classical and deep)
1% contamination
3% contamination
7% contamination
ours: rules + main detector
rules only
main detector: TCN-GAT, hierarchy, multi-scale
– learned graph (TCN, hierarchy, multi-scale)
– multi-scale (raw band only)
– hierarchy (flat residual)
– graph, – multi-scale (TCN, raw band)
main detector, NB3 event formation
mean(LightGBM, TCN-GAT)
LightGBM + learned-graph correction
LightGBM + TCN correction
LightGBM forecast
LightGBM, NB3 event formation
LightGBM nowcast (raw band)
STL + MAD
Isolation Forest
LOF
GDN (deep: learned graph)
USAD (deep: autoencoders)
0.0
0.2
0.4
0.6
0.8
1.0
event F1 (●) and step AUC-PR (□)